

A useful score must earn trust

Comparing logistic regression, XGBoost and TabICLv2
across performance, interpretability, stability and fairness

25,835 PEOPLE

Georgia parole cohort

3 model families

1 deployment
recommendation

Start with the decision, not the algorithm

- Prioritize limited, voluntary re-entry support at supervision start
- Estimate a recorded three-year arrest outcome — not inherent criminality
- Use only to expand help; never to justify sanctions, detention or surveillance
- Success means useful allocation without unacceptable subgroup harm

A clean test set and a strict time boundary

18,028

TRAIN

7,807

UNTOUCHED TEST

29

BASELINE FIELDS

57.4%

TEST TARGET RATE

- Post-release violations, tests, programs and employment are excluded to prevent leakage
- Race, gender and geography are excluded from scoring; race and gender remain in the audit layer
- The official NIJ split preserves an honest final evaluation

Three models expose different trade-offs

LOGISTIC

White-box anchor

Regularized linear score

Signed, inspectable effects

XGBOOST

Nonlinear workhorse

550 shallow trees

Fast operational scoring

TABICLv2

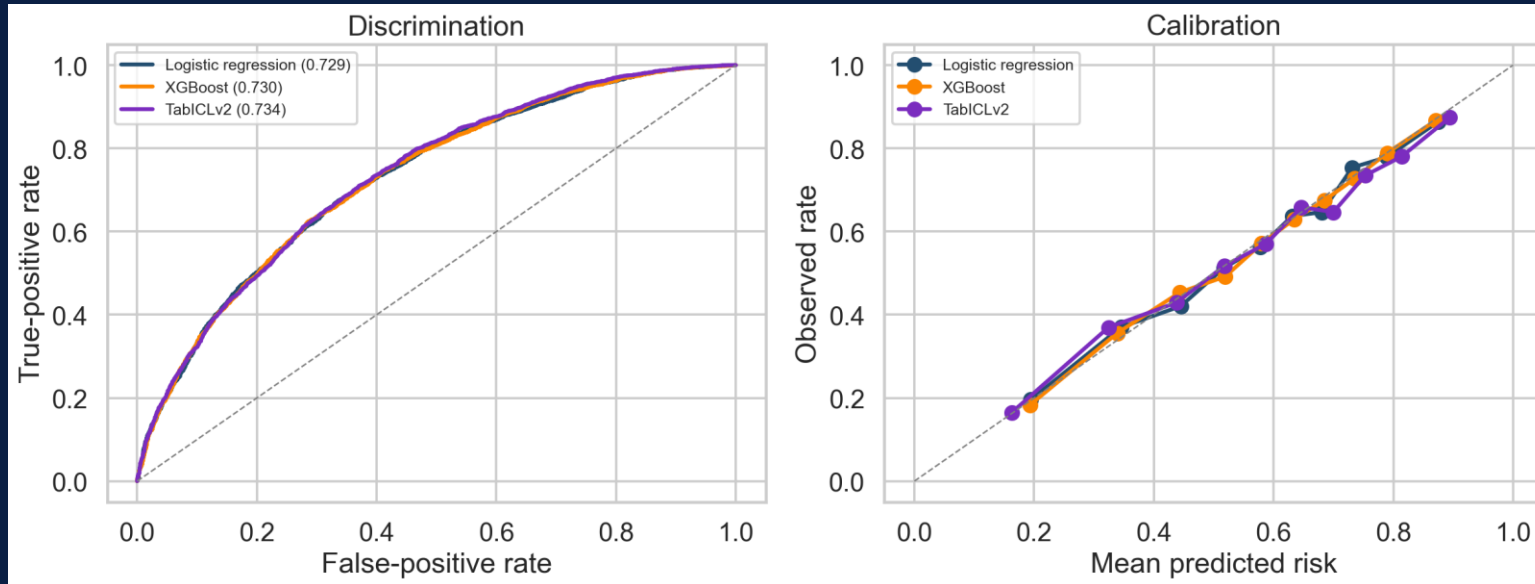
Foundation model

Pretrained in-context
transformer

Two ensemble views

Same eligible fields · Same held-out people · Same metrics

TabICLv2 wins narrowly; XGBoost calibrates best



0.2039

BEST BRIER · TABICL

0.7336

BEST AUC · TABICL

0.0118

BEST ECE · XGBOOST

Model value is a scenario, not a causal claim

20%

SERVICE CAPACITY

1,282–1,292

EVENTS CAPTURED

≈ 28.7%

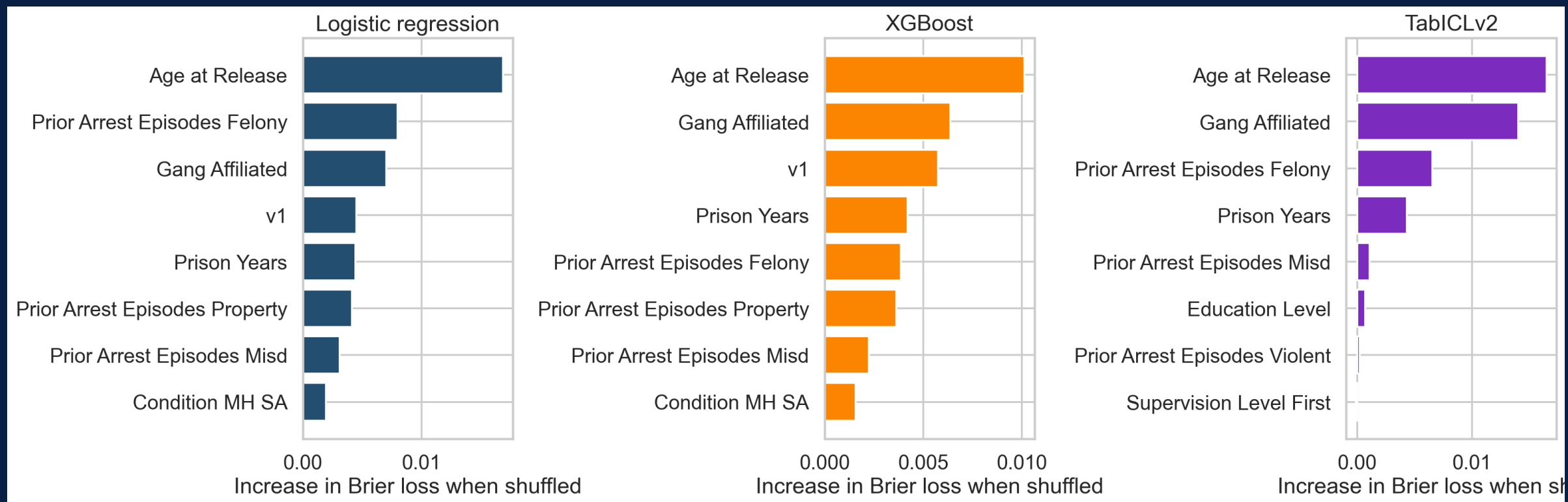
RECALL

\$5.0–5.1M

ILLUSTRATIVE NET

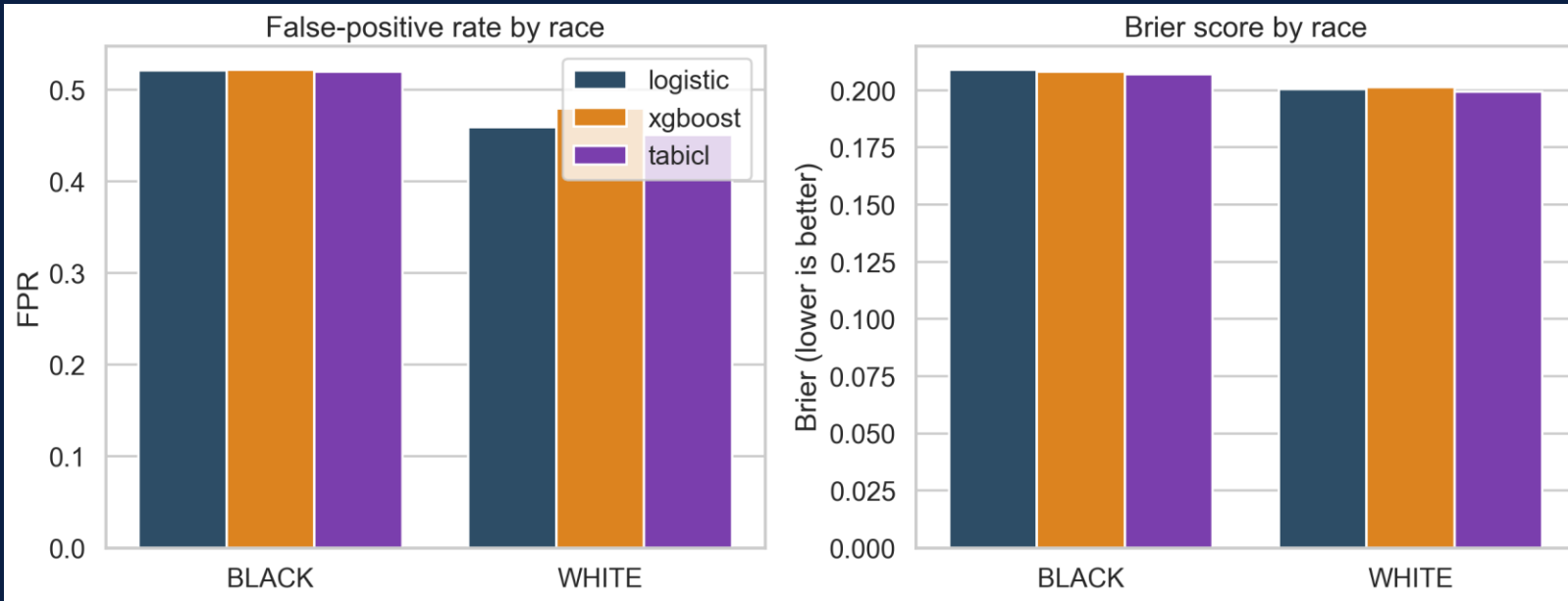
- \$5,000 support cost · \$50,000 event cost · 20% assumed effectiveness
- Every assumption is editable in the client application
- A randomized or strong quasi-experimental pilot must estimate actual intervention impact

The main drivers recur across model families



Permutation importance shows model reliance, not causal levers.

Fairness changes the preferred model

**0.042**

RACE FPR GAP · XGB

0.107

GENDER FPR GAP · XGB

TabICLv2 gender FPR gap

0.273

Sampling stability is similar; temporal stability is unknown

0.022–0.023

AUC CI WIDTH

0.0079–

0.0088
BRIER CI WIDTH

400

BOOTSTRAPS

Unavailable

TIME SPLIT

- Feature-destruction stress tests confirm material input reliance
- A later release cohort is a hard gate before operational launch
- Monitor calibration, input drift and subgroup gaps every quarter

Recommend XGBoost for a controlled pilot

WHY

Near-tied accuracy

Best calibration

Smallest observed race and gender FPR gaps

Fast operations

CONDITIONS

Benefit-only allocation

Logistic challenger

Prospective shadow validation

Appeal, override logs and stop rules

The application makes trade-offs testable

COMPARE

Performance and calibration

AUDIT

Race and gender metrics

SIMULATE

Costs, capacity and benefit

TEST

Held-out people and scenarios

```
streamlit run app.py
```

Deployment is a staged evidence program



Launch only if the score demonstrates benefit without unacceptable subgroup harm.

Sources: NIJ challenge brief and data; TabICLv2 model card and papers. Full methodology in the repository.